

Solutions HW#05

1.

$$(a) \quad \langle x, y \rangle = x^T \bar{y} = \sum_{i=1}^n x_i \bar{y}_i$$

$$(i) \quad \overline{\langle y, x \rangle} = \overline{y^T \bar{x}} = \overline{\sum_{i=1}^n y_i \bar{x}_i} = \sum_{i=1}^n \bar{y}_i x_i \\ = \langle x, y \rangle$$

$$(ii) \quad \langle \alpha_1 x_1 + \alpha_2 x_2, y \rangle = (\alpha_1 x_1 + \alpha_2 x_2)^T \bar{y} \\ = \alpha_1 x_1^T \bar{y} + \alpha_2 x_2^T \bar{y} \\ = \alpha_1 \langle x_1, y \rangle + \alpha_2 \langle x_2, y \rangle$$

$$(iii) \quad \langle x, x \rangle = \sum_{i=1}^n x_i \bar{x}_i = \sum_{i=1}^n |x_i|^2$$

Hence, $\langle x, x \rangle \geq 0$ for any $x \in \mathbb{C}^n$,

and $\langle x, x \rangle = 0 \Leftrightarrow |x_i|^2 = 0 \quad i=1, \dots, n,$

$\Rightarrow x = 0.$

$$(b) \langle x, y \rangle = \bar{x}^T y = \sum_{i=1}^n \bar{x}_i y_i$$

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$$(i) \overline{\langle y, x \rangle} = \overline{\bar{y}^T x} = \overline{\sum_{i=1}^n \bar{y}_i x_i}$$

$$= \sum_{i=1}^n y_i \bar{x}_i = \bar{x}^T y = \langle x, y \rangle$$

$$(ii) \langle x, \beta_1 y_1 + \beta_2 y_2 \rangle = \bar{x}^T (\beta_1 y_1 + \beta_2 y_2)$$

$$= \beta_1 \bar{x}^T y_1 + \beta_2 \bar{x}^T y_2 =$$

$$= \beta_1 \langle x, y_1 \rangle + \beta_2 \langle x, y_2 \rangle$$

$$(iii) \langle x, x \rangle = \bar{x}^T x = \sum_{i=1}^n \bar{x}_i x_i = \sum_{i=1}^n |x_i|^2$$

Hence, $\langle x, x \rangle \geq 0$ for any $x \in \mathbb{C}^n$,

and $\langle x, x \rangle = 0 \Leftrightarrow |x_i|^2 = 0 \quad i=1, \dots, n$

$\Rightarrow x = 0$.

$$2. \quad \langle f, g \rangle = \int_{-1}^1 f(x)g(x) dx$$

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$$p_0(x) = 1$$

$$p_2(x) = \frac{1}{2} (3x^2 - 1)$$

$$p_1(x) = x$$

$$p_3(x) = \frac{1}{2} (5x^3 - 3x)$$

$$\langle p_0, p_3 \rangle = \int_{-1}^1 p_0(x) p_3(x) dx$$

$$= \int_{-1}^1 \frac{1}{2} (5x^3 - 3x) dx$$

$$= \frac{1}{2} \left[\frac{5x^4}{4} - \frac{3x^2}{2} \right] \Big|_{-1}^1$$

$$= \frac{1}{2} \left[\left(\frac{5}{4} - \frac{3}{2} \right) - \left(\frac{5}{4} - \frac{3}{2} \right) \right]$$

$$= 0$$

$$\langle p_1, p_2 \rangle = \int_{-1}^1 p_1(x) p_2(x) dx$$

$$= \int_{-1}^1 x \frac{1}{2} (3x^2 - 1) dx$$

$$= \int_{-1}^1 \frac{1}{2} (3x^3 - x) dx$$

$$= \frac{1}{2} \left[\frac{3}{4} x^4 - \frac{x^2}{2} \right] \Big|_{-1}^1$$

$$= \frac{1}{2} \left[\left(\frac{3}{4} - \frac{1}{4} \right) - \left(\frac{3}{4} - \frac{1}{2} \right) \right]$$

$$= 0.$$

$$3. \quad v' = y' = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

$$\|v'\|^2 = 6$$

$$v^2 = y^2 - a_{21} v'$$

$$a_{21} = \frac{\langle y^2, v' \rangle}{\|v'\|^2} = \frac{[4 \quad 0 \quad -1] \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}}{6}$$

$$= \frac{3}{6} = \frac{1}{2}$$

$$\therefore v^2 = \begin{bmatrix} 4 \\ 0 \\ -1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} = \begin{bmatrix} 3\frac{1}{2} \\ 1 \\ -1\frac{1}{2} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 7 \\ 2 \\ -3 \end{bmatrix}$$

$$\cancel{v^2} \quad \|v^2\|^2 = \frac{1}{4} (49 + 4 + 9) = \frac{62}{4} = \frac{31}{2}$$

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$$V^3 = y^3 - a_{31}V^1 - a_{32}V^2$$

$$a_{31} = \frac{\langle y^3, V^1 \rangle}{\|V^1\|^2} = \frac{[-2 \ 2 \ 3] \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}}{6}$$

$$= \frac{-3}{6} = -\frac{1}{2}$$

$$a_{32} = \frac{\langle y^3, V^2 \rangle}{\|V^2\|^2} = \frac{[-2 \ 2 \ 3] \cdot \begin{bmatrix} 7 \\ 2 \\ -3 \end{bmatrix} (\frac{1}{2})}{3\frac{1}{2}}$$

$$= \frac{-19}{31}$$

$$V^3 = \begin{bmatrix} -2 \\ 2 \\ 3 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} + \frac{19}{31} \begin{bmatrix} 3\frac{1}{2} \\ 1 \\ -1\frac{1}{2} \end{bmatrix}$$

$$= \begin{bmatrix} 40 \\ 100 \\ 160 \end{bmatrix} \left(\frac{1}{62}\right) \approx \begin{bmatrix} 0.65 \\ 1.61 \\ 2.58 \end{bmatrix}$$

4.

(a) If A^{-1} , C^{-1} and $(C^{-1} + DA^{-1}B)^{-1}$ each exist, then

$$M = A^{-1} - A^{-1}B(C^{-1} + DA^{-1}B)^{-1}DA^{-1} \quad (*)$$

is well defined (means each of the terms exists and the indicated matrix products are compatible).

To show: The proposed inverse (*)

when multiplied by $(A + BCD)$

gives the identity.

We will ^{show} $M(A + BCD) = I$

$$\begin{aligned}
M \cdot (A + BCD) &= \\
&= I - A^{-1}B(C^{-1} + DA^{-1}B)^{-1}D + \\
&\quad + A^{-1}BCD - A^{-1}B(C^{-1} + DA^{-1}B)^{-1}DA^{-1}BCD \\
&= I - A^{-1}B \left[(C^{-1} + DA^{-1}B)^{-1}C^{-1} - I + (C^{-1} + DA^{-1}B)^{-1}DA^{-1}B \right] \cdot CD \\
&= I - A^{-1}B \left[(C^{-1} + DA^{-1}B)^{-1}(C^{-1} + DA^{-1}B) - I \right] CD \\
&= I - A^{-1}B [I - I]D \\
&= I
\end{aligned}$$

$$A^{-1} = \text{diag}([1, 2, 2, 1, 2])$$

$$C^{-1} = 5$$

~~$$C^{-1} + DA^{-1}B = 5 + 27$$~~

$$A^{-1}B = \begin{bmatrix} 1 \\ 0 \\ 4 \\ 1 \\ 6 \end{bmatrix}$$

$$DA^{-1}B = [1 \ 0 \ 2 \ 0 \ 3] \begin{bmatrix} 1 \\ 0 \\ 4 \\ 1 \\ 6 \end{bmatrix} = 27$$

$$C^{-1} + DA^{-1}B = 32 \Rightarrow (C^{-1} + DA^{-1}B)^{-1} = \frac{1}{32}$$

~~$$DA^{-1} = [1 \ 0 \ 8 \ 0 \ 12]$$~~

$$\therefore (A + BCD)^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \\ 4 \\ 1 \\ 6 \end{bmatrix} \left(\frac{1}{32} \right) [1 \ 0 \ 8 \ 0 \ 12]$$

$$= \begin{bmatrix} 1 & 0 & 0 & 6 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 6 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 6 & 0 & 0 & 2 \end{bmatrix} - \frac{1}{32} \begin{bmatrix} 1 & 0 & 8 & 0 & 12 \\ 0 & 0 & 0 & 0 & 0 \\ 4 & 0 & 32 & 0 & 48 \\ 6 & 0 & 48 & 0 & 72 \end{bmatrix}$$

$$= \frac{1}{32} \begin{bmatrix} 31 & 0 & -8 & 0 & -12 \\ 0 & 64 & 0 & 0 & 0 \\ -4 & 0 & 32 & 0 & -48 \\ -6 & 0 & -48 & 0 & -8 \end{bmatrix}$$

5. }
6. } See latex document. It
needs figures inserted.

7. We apply the normal equations.

$$\hat{x} = \alpha_1 y^1 + \alpha_2 y^2$$

where $G^T \alpha = b$ and

$$G = \begin{bmatrix} \langle y^1, y^1 \rangle & \langle y^2, y^1 \rangle \\ \langle y^1, y^2 \rangle & \langle y^2, y^2 \rangle \end{bmatrix}$$

$$b = \begin{bmatrix} \langle x, y^1 \rangle \\ \langle x, y^2 \rangle \end{bmatrix}$$

Doing the calculations, we have

~~$$\langle y^1, y^2 \rangle = \text{tr} \left(\begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \right) = \text{tr} \left(\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} \right) = 3$$~~

~~$$\langle y^1, y^2 \rangle = \langle y^2, y^1 \rangle = \text{tr}$$~~

~~$$\langle y^1, y^1 \rangle = \text{tr} \left(\begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix} \right) = \text{tr} \left(\begin{bmatrix} 1 & 0 \\ 4 & 0 \end{bmatrix} \right) = 1$$~~

$$\langle y^1, y^1 \rangle = \text{tr} \left(\begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix} \right) = \text{tr} \left(\begin{bmatrix} 5 & 0 \\ 0 & 0 \end{bmatrix} \right) = 5$$

$$\langle y^1, y^2 \rangle = \langle y^2, y^1 \rangle = \text{tr} \left(\begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \right) = \text{tr} \left(\begin{bmatrix} 3 & 3 \\ 0 & 0 \end{bmatrix} \right) = 3$$

$$\langle y^2, y^2 \rangle = \text{tr} \left(\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \right) = \text{tr} \left(\begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} \right) = 4$$

$$\langle x, y^1 \rangle = \text{tr} \left(\begin{bmatrix} 0 & 2 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix} \right) = \text{tr} \left(\begin{bmatrix} 4 & 0 \\ -1 & 0 \end{bmatrix} \right) = 4$$

$$\langle x, y^2 \rangle = \text{tr} \left(\begin{bmatrix} 0 & 2 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \right) = \text{tr} \left(\begin{bmatrix} 2 & 2 \\ -1 & -1 \end{bmatrix} \right) = 1$$

$$\hat{0}_0 \quad \begin{bmatrix} 5 & 3 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

$$\hat{0}_0 \quad \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{11} \begin{bmatrix} 4 & -3 \\ -3 & 5 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \end{bmatrix} = \frac{1}{11} \begin{bmatrix} 13 \\ -7 \end{bmatrix} = \begin{bmatrix} 1.18 \\ -0.64 \end{bmatrix}$$

$$\hat{X} = \frac{13}{11} \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix} - \frac{7}{11} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} \frac{6}{11} & -\frac{7}{11} \\ \frac{19}{11} & -\frac{7}{11} \end{bmatrix}$$

8. Let $\gamma := d(x, M)$, and suppose that $m_1, m_2 \in M$ satisfy $\|x - m_i\| = \gamma$.

To show $m_1 = m_2$ when the norm is strict.

Because M is a subspace, $\frac{m_1 + m_2}{2} \in M$.

Hence,

$$\begin{aligned}
 \gamma &= \inf_{y \in M} \|x - y\| \leq \|x - \frac{m_1 + m_2}{2}\| \\
 &= \left\| \frac{x - m_1}{2} + \frac{x - m_2}{2} \right\| \\
 &\leq \frac{1}{2} \|x - m_1\| + \frac{1}{2} \|x - m_2\| \\
 &= \frac{\gamma}{2} + \frac{\gamma}{2} \\
 &= \gamma.
 \end{aligned}$$

Hence, $\|(x - m_1) + (x - m_2)\| = \|x - m_1\| + \|x - m_2\|$

Because the norm is strict, $\exists \alpha \geq 0$
 such that either (i) $(x-m_1) = \alpha(x-m_2)$ or
 (ii) $(x-m_2) = \alpha(x-m_1)$.

In either case, we deduce from
 $\gamma = \|x-m_1\| = \|x-m_2\|$, that $\gamma = \alpha\gamma$,
 and, ~~thus~~ because $\gamma \neq 0$, we have $\alpha = 1$.

With $\alpha = 1$, we have $x-m_1 = x-m_2$,
 and thus $m_1 = m_2$.

□

9.

(a) Not strictly normed. Let

$$x = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ and } y = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

$$\text{Then } \|x+y\|_1 = 2 = \|x\|_1 + \|y\|_1,$$

but there does not exist any $\alpha \geq 0$
such that ~~either~~ $x = \alpha y$ or $y = \alpha x$.

(c) Not strictly normed. Let

$$x = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \text{ and } y = \begin{bmatrix} 4 \\ 1 \end{bmatrix}.$$

$$\text{Then } 6 = \|x+y\|_\infty = \|x\|_\infty + \|y\|_\infty,$$

but there does not exist any $\alpha \geq 0$
such that either $x = \alpha y$ or $y = \alpha x$.

(b) Strictly normed The result is true for any norm induced by an inner product. Hence we give the proof for $\|x\| = \langle x, x \rangle^{1/2}$.

Let $x, y \in X$

Case 1 Either x or y is zero. Then

$\|x+y\| = \|x+y\|$ is always true and either $x=0 \cdot y$ or $y=0 \cdot x$ holds. $\square$

Case 2 Both $x \neq 0$ and $y \neq 0$, but

$\{x, y\}$ is linearly dependent. Then ~~suppose~~ $x = \alpha y$ ^{for some $\alpha \in \mathbb{R}$} . It follows that

$$\|x+y\| = \|1+\alpha\| \cdot \|y\|$$

and

$$\|x\| + \|y\| = (1 + |\alpha|) \|y\|.$$

Because $y \neq 0$, we have $\|x+y\| = \|x\| + \|y\| \Leftrightarrow$

$$|1 + \alpha| = 1 + |\alpha|. \text{ and } \Leftrightarrow \alpha \geq 0.$$

~~deduce that~~ $\alpha \geq 0$. Hence,

Hence, $\|x+y\| = \|x\| + \|y\| \Leftrightarrow x = \alpha y, \alpha \geq 0.$

□

~~The~~

Case 3 $\{x, y\}$ is linearly independent.

By the Gram-Schmidt procedure,
there exists $v \in X$ such that

$$x \perp v \text{ and } \text{span}\{x, y\} = \text{span}\{x, v\}.$$

Write $y = \alpha_1 x + \alpha_2 v$, so that

$$x+y = (1+\alpha_1)x + \alpha_2 v$$

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$$\|x+y\| = \|x\| + \|y\| \quad \text{if, and only if}$$

$$\|x+y\|^2 = \|x\|^2 + \|y\|^2 + 2\|x\| \cdot \|y\|$$

By the Pythagorean Theorem,

$$\begin{aligned}\|x+y\|^2 &= \|(1+\alpha_1)x + \alpha_2v\|^2 \\ &= (1+\alpha_1)^2 \|x\|^2 + (\alpha_2)^2 \|v\|^2 \\ &= [1 + 2\alpha_1 + (\alpha_1)^2] \|x\|^2 + (\alpha_2)^2 \|v\|^2\end{aligned}$$

Furthermore

$$\|y\|^2 = (\alpha_1)^2 \|x\|^2 + (\alpha_2)^2 \|v\|^2$$

$$\text{Hence, } \|x+y\|^2 = \|x\|^2 + \|y\|^2 + 2\|x\| \cdot \|y\|$$

if, and only if,

$$2\alpha_1 \|x\|^2 = 2\|x\| \cdot \|y\|$$

Because $\|x\| \neq 0$ from $\{x, y\}$ linearly independent, we have

$$\alpha_1 \|x\| = \|y\|$$

so $\alpha_1 \geq 0$. Moreover

$$(\alpha_1)^2 \|x\|^2 = \|y\|^2 = (\alpha_1)^2 \|x\|^2 + (\alpha_2)^2 \|v\|^2$$

if, and only if, $\alpha_2 = 0$.

Hence, $\|x+y\| = \|x\| + \|y\|$ if, and

only if $y = \alpha_1 x$, $\alpha_1 \geq 0$, and

~~the~~

□